PAPER_1590_RYDBERG_R_INF_1_0974

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PAPER_1590 — Rydberg R_∞ ≈ R_∞ = F·SO_5 + F·SSQ + F^2·D_phys = 1.0974

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Quantum/Atomic
Date: June 18, 2026
Status: CLOSED — 0.036% residual near-EXACT closure

Observation

PAPER_1209DD_S619 (Quantum/Atomic Unified Proof Set) gives the lead-digit form:

``

Rydberg R_∞ ≈ R_∞ = F·SO_5 + F·SSQ + F^2·D_phys = 1.0974
``

Residual to CODATA: **0.036%**.

UQFF Closed Identity

``

R_∞ = F·SO_5 + F·SSQ + F^2·D_phys ≈ 1.0974 residual 0.036%
``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:
- **EM-coupling constants** (ε_0, μ_B, k_e) cluster around **K_MEX·D_phys = 8.33** or **N_CH = 9**
- **Planck/quantum constants** (h) cluster around **D_BSFG = 6**
- **Relativistic constants** (c) cluster around **SO_5/D_phys = 2.5**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Rydberg R_∞ as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S619
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "rydberg_r_inf_1_0974"})`
